Lesson 12.1 - The Chi Square Test

> **Video:** Chi Square Test by The Organic Chemistry Tutor **Chapter:** 12 - The Chi-Square Distribution

The big picture: testing counts, not averages

Chapters 10 and 11 tested **one number**: a mean, a proportion, or a difference. Many questions are about **how counts are spread across several categories**: absences on each weekday, M&M colors in a bag, votes for 4 candidates. There is no single average to test. We need a way to compare a **whole table of counts** with what H_0 predicts.

1. H_0 gives the expected counts. H_0 says what share each category should get. Multiply by the sample size:

$$E = n \times p$$

With 100 teachers and "absences are spread equally over 5 days", each day expects $E = 100 \times \frac{1}{5} = 20$.

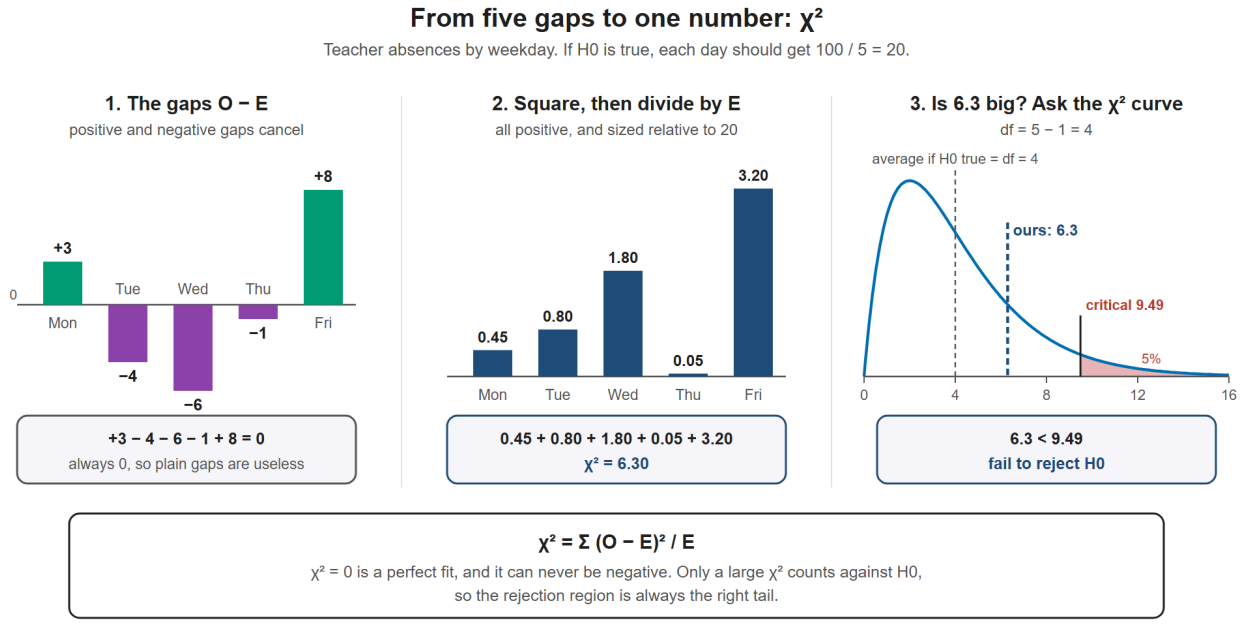
2. Why not just add up the gaps $O - E$? Because they always add up to 0. The observed and expected counts both total 100, so every extra absence on one day is a missing one somewhere else: $+3 - 4 - 6 - 1 + 8 = 0$.

3. Square, then divide by E.

- **Squaring** makes every gap positive, so the gaps can't cancel. It also gives big gaps extra weight.
- **Dividing by E** measures each gap **relative to what was expected**. Missing by 8 is a lot when you expect 20, but almost nothing when you expect 1000.

Add up the pieces for every category, and you get one number for the total mismatch:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$



The five gaps cancel out, so they are squared and scaled by E before being added. The total, 6.3, is then compared with the χ^2 curve to see whether it is unusually large.

4. Why the test is always right-tailed. χ^2 can't be negative, and $\chi^2 = 0$ means a perfect fit. Only a **large** χ^2 is evidence against H_0, so the rejection region is always the right tail. A χ^2 near 0 just means the data fit very well.

5. Where the χ^2 curve comes from. Each piece $\frac{O-E}{\sqrt{E}}$ behaves roughly like a z-score, so χ^2 is roughly a **sum of squared z-scores**. That sum has its own distribution, the **chi-square distribution**:

- It starts at 0 and is skewed right, because squares are never negative.
- Its shape depends on the **degrees of freedom**. On average, χ^2 is about **df** when H_0 is true. Here df = 4 and we got 6.3, a bit more than average but not unusual.

6. Why df $= k - 1$. The counts must add up to the total. Once you know 4 of the 5 days, the 5th is forced: $100 - 23 - 16 - 14 - 19 = 28$. Only $k - 1$ counts are free to vary.

The 5 steps are the same as in Chapters 10 and 11:

1. Write H_0 (the expected distribution) and H_a (it doesn't fit).
2. Choose α.
3. Find the critical value from the χ^2 table, with df $= k - 1$ and right-tail area α.
4. **Compute** $\chi^2 = \sum(O - E)^2/E$. ← this lesson
5. χ^2 greater than the critical value → reject H_0. Otherwise → fail to reject H_0.

The goodness-of-fit test

Does the **observed** distribution of counts across categories match an **expected** distribution?

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

- O = observed count, E = expected count, for each category
- **Degrees of freedom** = $k - 1$, where k = number of categories
- It is always a **right-tailed** test: a large χ^2 means the observed counts are far from the expected ones

The chi-square distribution is **not symmetric**; it is skewed to the right. The critical value χ^2_α cuts off area α in the right tail:

- calculated χ^2 > critical value → **rejection region** → reject H_0
- calculated χ^2 ≤ critical value → **do-not-reject region**

Example - which weekday has the most absences?

A principal expects absences to be spread **equally** over the 5-day school week. 100 teachers were asked which day had the most absences. Test at $\alpha = 0.05$.

Day	Mon	Tue	Wed	Thu	Fri	Total
Observed O	23	16	14	19	28	100
Expected E	20	20	20	20	20	100

Expected counts: $E = n \times p = 100 \times \frac{1}{5} = 20$ for each day.

Hypotheses:

- H_0: absences happen with **equal frequencies** (a uniform distribution)
- H_a: absences do **not** happen with equal frequencies

Critical value: df $= k - 1 = 5 - 1 = 4$. In the χ^2 table with right-tail area 0.05: $\chi^2_{0.05,\,4} = 9.49$.

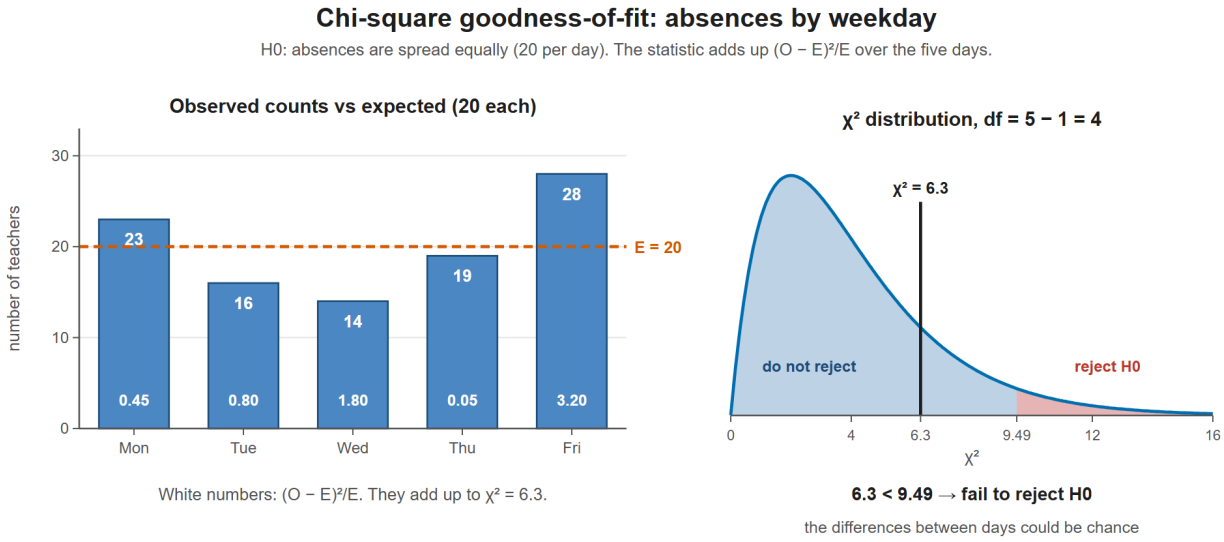
Test statistic:

Day	$O - E$	$(O - E)^2$	$(O - E)^2/E$
Mon	3	9	0.45
Tue	−4	16	0.80
Wed	−6	36	1.80
Thu	−1	1	0.05
Fri	8	64	3.20
Sum		**126**	**6.30**

$$\chi^2 = \sum \frac{(O - E)^2}{E} = \frac{9 + 16 + 36 + 1 + 64}{20} = \frac{126}{20} = 6.3$$

(Every E is 20 here, so you can add the squares first and divide once.)

$6.3 < 9.49$, which is in the do-not-reject region. **Fail to reject H_0.** The variation between days is not large enough to say that absences are unevenly spread across the week.



Left: Friday and Wednesday are furthest from the expected 20, so they contribute the most to χ^2. Right: the total, 6.3, is still below the critical value 9.49, so the uneven pattern could be due to chance.

Key takeaways

- The gaps $O - E$ always add up to 0, so they are squared and divided by E before being added.

- $\chi^2 = \sum(O - E)^2/E$ measures how far the observed counts are from the expected ones.
- Goodness of fit: df = number of categories − 1. It is always right-tailed.
- Large χ^2 (beyond the critical value) → reject H_0 (the data don't fit the expected distribution).